

Ja Young Choi

Postdoctoral Associate
Department of Speech, Language & Hearing Sciences, Boston University
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Education

- 2022 **Ph.D. in Speech and Hearing Bioscience and Technology**
Harvard University, Cambridge, MA
Dissertation: *The cognitive and neural bases of processing talker variability in speech perception*
- 2014 **B.A. in Cognitive Science, Linguistics, and Psychology**
Rice University, Houston, TX

Research experience

- 2025– **Postdoctoral Scholar**
Stepp Lab for Sensorimotor Rehabilitation Engineering, Boston University, Boston, MA
PI: Cara Stepp, Ph.D.
- 2023–2025 **Postdoctoral Scholar**
SoundBrain Lab, Northwestern University, Evanston, IL
PI: Bharath Chandrasekaran, Ph.D.
- 2016–2022 **Ph.D. Student**
Communication Neuroscience Research Lab, Boston University, Boston, MA
PI: Tyler Perrachione, Ph.D.
- 2014–2016 **Research Analyst**
Communication Neuroscience Research Lab, Boston University, Boston, MA
PI: Tyler Perrachione, Ph.D.
- 2014 **Undergraduate Research Assistant**
Child Language Acquisition Lab, Rice University, Houston, TX
PI: Özge Gurcanlı, Ph.D.
- 2012–2014 **Undergraduate Research Assistant**
Schnur Lab, Rice University, Houston, TX
PI: Tatiana Schnur, Ph.D.

Research funding

- 2026–2027 **New Investigators Research Grant (Role: PI)** Amount: \$10,000
American Speech-Language-Hearing Foundation
Project Title: Auditory-perceptual contributions to monopitch speech in Parkinson's disease
- 2026–2027 **Student Research Grant – Postdoc (Role: PI)** Amount: \$5,000
Sargent College of Rehabilitation Sciences, Boston University
Project Title: Auditory Perceptual factors Contributing to Monopitch Speech in Parkinson's Disease

Submitted & pending grant

2027–2030 **NIH R21 DC024784 (Role: PI)** Requested: \$613,125.00
NIH National Institute of Deafness and Other Communication Disorders
Project Title: Monopitch speech in Parkinson's disease: linking perception, production and treatment response

Honors & awards

2025 Postdoctoral Abstract Merit Award – Honorable Mention for *17th Annual Meeting of the Society for the Neurobiology of Language* 2025
2020 Best Student Paper in Speech Communication – First place for *179th Meeting of the Acoustical Society of America* 2020
2019 International Phonetic Association Student Award for *19th International Congress of Phonetic Sciences* 2019
2016–2020 Kwanjeong Educational Scholarship
2012 Center for the Study of Languages Study Abroad Scholarship
2012 Houston Junior Chamber of Commerce Award: Study in China Scholarship
2011–2014 President's Honor Roll, Rice University

Publications

Choi, J. Y., Edelstein, I., Abur, D., & Stepp, C.E. (submitted). Fundamental frequency changes at syntactic boundaries and listener perception of monopitch in Parkinson's disease.
Choi, J. Y., Dragicovic, D.A., Li, S.H., Davis, R.A., Siedman, K.A., Toglia, D., Dunsmuir, C.J., Rojas, J.M., Estrada, A., Balz, M.A., Perkell, J.S., & Stepp, C.E. (in revision). Articulatory changes during clear speech in people with Parkinson's disease.
Choi, J. Y., Xiong, S., McHaney, J. R., & Chandrasekaran, B. (2026). Reverse hierarchical processing of speech in talker identification. *European Journal of Neuroscience*, 63(8), e70520.
Liu, Y., **Choi, J. Y.**, & Perrachione, T.K. (2025). Systematic bias in surface area asymmetry measurements from automatic cortical parcellations. *Brain Structure and Function*, 230, 126.
Choi, J. Y., Kou, R.S.N., & Perrachione, T.K. (2022). Distinct mechanisms for talker adaptation operate in parallel on different timescales. *Psychonomic Bulletin & Review*, 29, 627-634.
Choi, J. Y., & Perrachione, T.K. (2019). Noninvasive neurostimulation of left temporal lobe disrupts rapid talker adaptation in speech processing. *Brain and Language*, 196, 104655.
Choi, J. Y., & Perrachione, T.K. (2019). Time and information in perceptual adaptation to speech. *Cognition*, 192, 103982.
Choi, J. Y., Hu, E.R., & Perrachione, T.K. (2018). Varying acoustic-phonemic ambiguity reveals that talker normalization is obligatory in speech processing. *Attention, Perception, & Psychophysics*, 80(3), 784-797.

Peer-reviewed conference proceedings

Choi, J. Y., & Perrachione, T.K. (2019). Rapid adaptation to talker-specific phonetic detail is disrupted by noninvasive brain stimulation. *19th International Congress of Phonetic Sciences* (Melbourne, August 2019).

Work in prep

Choi, J. Y., Dragicevic, D. A., Vojtech, J., & Stepp, C. E. (in prep). Quantifying metrics that account for listener perceptions of articulatory imprecision in people with Parkinson's disease.

Choi, J. Y., Guo, Z.-C., Parker, A., Chandrasekaran, B., & Hampton-Wray, A. (in prep). Neural representations of speech sounds in children who stutter.

Conference presentations

Perrachione, T.K., **Choi, J. Y.**, Moore, A.A., Belisle, R.M., & Lim, S.-J. (2026). Cortical networks associated with talker-specific speech processing. *3rd Interdisciplinary Conference on Voice Identity (VoicelD): Perception, Production, and Computational Approaches (Jena, Germany, September 2026)*.

Wittenback, H., Corbin, N., Sitek, K., **Choi, J. Y.**, Henry, M., & Wiseman, K. (2026). Effect of talker similarity and talker language on children's ability to identify talkers. *American Auditory Society Scientific and Technology Meeting (Scottsdale, Feb 2026)*.

Wiseman, K. B., Corbin, N., Wittenback, H., Sitek, K. R., **Choi, J. Y.**, & Chandrasekaran, B. (2026). Talker identification in children with typical hearing, hearing aids, and cochlear implants. *Conference on Cochlear Implants (CI2026 Chicago, May 2026)*.

Choi, J. Y., Guo, Z.-C., Chandrasekaran, B., & Hampton Wray, A. (2025). Neural representations of speech sounds in children who stutter. *17th Annual Meeting of the Society for the Neurobiology of Language (Washington DC, September 2025)*.

Choi, J. Y., Gnanateja, G. N., Basavanahalli Jagadeesh, A., & Chandrasekaran, B. (2025). Revisiting Laurel vs. Yanny: insights from an online experiment and an FFR experiment. *17th Annual Meeting of the Society for the Neurobiology of Language (Washington DC, September 2025)*.

Choi, J. Y., Sitek, K. R., Langlois, L.-P., Deschamps, S., Chen, J.-K., Roark, C., Baum, S., Chandrasekaran, B., & Klein, D. (2025). Behavioral identification and neural representations of talkers' voices vary with lifelong language experience. *17th Annual Meeting of the Society for the Neurobiology of Language (Washington DC, September 2025)*.

Choi, J. Y., Xiong, S., McHaney, J. R., & Chandrasekaran, B. (2024). Pupillary measures of identifying talkers in native language and unfamiliar language. *Advances and Perspectives in Auditory Neuroscience XXII (Chicago, October 2024)*.

Liu, Y., **Choi, J. Y.**, & Perrachione, T.K. (2024). Hemispheric biases in automatic cortical parcellations exaggerate surface area lateralization in primary auditory cortex and other key language areas. *Advances and Perspectives in Auditory Neuroscience XXII (Chicago, October 2024)*.

Basavanahalli Jagadeesh, A., **Choi, J. Y.**, Gnanateja, G.N., & Chandrasekaran, B. (2024). Laurel vs Yanny – An investigation into the neural mechanisms of an Auditory Illusion. *5th Frequency Following Response Workshop (Chicago, June 2024)*.

Choi, J. Y. & Perrachione, T.K. (2023). Functional and structural connectivity of auditory areas that process talker variability in speech. *15th Annual Meeting of the Society for the Neurobiology of Language (Marseille, October 2023)*.

Perrachione, T.K. & **Choi, J. Y.** (2022). Hemispheric asymmetries in the cortical myeloarchitecture parallel the functional lateralization of language. *14th Annual Meeting of the Society for the Neurobiology of Language (Philadelphia, October 2022)*.

Lee, J.J., Scott, T.L., Carter, Y.D., **Choi, J. Y.**, & Perrachione, T.K. (2022). Functional selectivity and structural connectivity of the cortical language network are intact in dyslexia. *29th Annual Meeting of the Cognitive Neuroscience Society (San Francisco, April 2022)*.

- Lee, J.J., Scott, T.L., Carter, Y., **Choi, J. Y.**, & Perrachione, T.K. (2022). Cortical language network functional neuroanatomy in dyslexia. *Neurobiology of Language: Key Issues and Ways Forward II (Max Planck Institute & Virtual, March 2022)*.
- Choi, J. Y.**, Kou, R.S.N., & Perrachione, T.K. (2020). Parametrically varying speech adapter length suggests two mechanisms for talker adaptation. *179th Meeting of the Acoustical Society of America (Online, Dec 2020)*.
- Choi, J. Y.**, Torre, G.A., Carter, Y.D., Scott, T.L., Ghosh, S.S., & Perrachione, T.K. (2020). Multivoxel pattern analyses of brain structure to classify dyslexia. *27th Annual Meeting of the Cognitive Neuroscience Society (Boston, March 2020)*.
- Torre, G.A., **Choi, J. Y.**, Scott, T.L., Carter, Y.D., & Perrachione, T.K. (2020). Differences in left fusiform gyrus morphometry in adults with dyslexia: Voxel- and surface-based analyses. *27th Annual Meeting of the Cognitive Neuroscience Society (Boston, March 2020)*.
- Choi, J. Y.** & Perrachione, T.K. (2019). Noninvasive neurostimulation reveals a causal role for left superior temporal lobe in speech adaptation. *11th Annual Meeting of the Society for the Neurobiology of Language (Helsinki, August 2019)*.
- Scott, T.L., Carter, Y.D., **Choi, J. Y.**, & Perrachione, T.K. (2019). Relationships between phonological working memory and language processing in adults with dyslexia. *11th Annual Meeting of the Society for the Neurobiology of Language (Helsinki, August 2019)*.
- Scott, T.L., Dougherty, S.C., **Choi, J. Y.**, & Perrachione, T.K. (2018). Nonword repetition recruits distinct and overlapping nodes of language and working memory networks. *10th Annual Meeting of the Society for the Neurobiology of Language (Quebec City, August 2018)*.
- Scott, T.L., Dougherty, S.C., **Choi, J. Y.**, & Perrachione, T.K. (2018). Common recruitment of neural resources for phonological working memory regardless of behavioral demands. *25th Annual Meeting of the Cognitive Neuroscience Society (Boston, April 2018)*.
- Scott, T.L., Dougherty, S.C., **Choi, J. Y.**, & Perrachione, T.K. (2017). The role of language-specific vs. domain general systems in phonological working memory. *23rd Annual Meeting of the Organization for Human Brain Mapping (Vancouver, June 2017)*.
- Perrachione, T.K., Dougherty, S.C., **Choi, J. Y.**, & Hu, E.R. (2016). Noninvasive brain stimulation to facilitate foreign language speech-sound learning in low-aptitude learners. *172nd meeting of the Acoustical Society of America (Honolulu, November 2016)*.
- Choi, J. Y.**, Hu, E.R., & Perrachione, T.K. (2016). Orthogonal interference of indexical information occurs even when phonetic contrasts are unambiguous across talkers. *171st meeting of the Acoustical Society of America (Salt Lake City, May 2016)*.
- Perrachione, T.K. & **Choi, J. Y.** (2016). Extrinsic talker normalization via rapid accumulation of talker-specific phonetic detail. *171st meeting of the Acoustical Society of America (Salt Lake City, May 2016)*.
- Choi, J. Y.**, Minas, J.E., Finn, A.S., Gabrieli, J.D.E., & Perrachione, T.K. (2015). Functional brain changes associated with learning a novel phonological contrast. *21st Annual Meeting of the Organization for Human Brain Mapping (Honolulu, June 2015)*.
- Choi, J. Y.**, Wei, T., & Schnur, T.T. (2014). Effect of cognitive control on semantic processing. *Rice University Undergraduate Research Symposium (Houston, April 2014)*.

Teaching experience

Fall 2026 **Instructor of record**
Phonetics (SAR SH 221)
 Department of Speech, Hearing and Language Sciences, Boston University

- Spring 2025 **Guest lecturer**
Interdisciplinary Topics in Communication Arts & Sciences (CMN 101-0)
School of Communication, Northwestern University
- Fall 2021 **Laboratory instructor**
Neurobiology of Language (SAR SH 501)
Department of Speech, Hearing and Language Sciences, Boston University

Mentorship

- 2026 **Maxwell Murphy**
Undergraduate Student, Neuroscience, Boston University
• Honors neuroscience thesis
- 2025 **Jacqueline Martinez**
Undergraduate Student, Neuroscience, Northwestern University
• Summer internship grant recipient
- 2025 **Jaimie Hong**
Undergraduate Student, Computer Science & Biology, Northwestern University
• Recipient of 2025 Spring Academic Year Undergraduate Research Grant
(co-sponsored with Dr. Bharath Chandrasekaran)
- 2024–2025 **Louis-Phillipe Langlois**
MSc student, Neuroscience, McGill University
- 2023–2025 **Shengyue Xiong**
PhD student, Communication Sciences and Disorders, Northwestern University
- 2018–2019 **Rita S. N. Kou**
MS student, Speech-Language Pathology, Boston University

Service

Ad hoc journal reviews

Brain and Language
Communications Biology
European Journal of Neuroscience
Psychonomic Bulletin & Review

Outreach

Judge, Undergraduate Research and Arts Exposition, Northwestern University